

DiskJoin: Large-scale Vector Similarity Join with SSD

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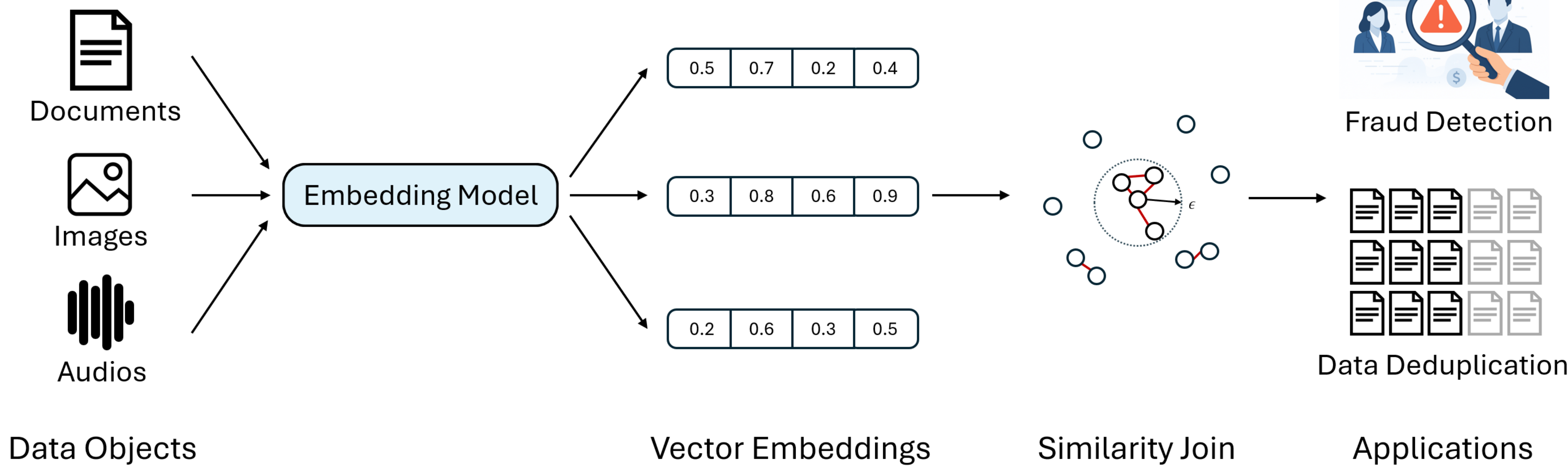
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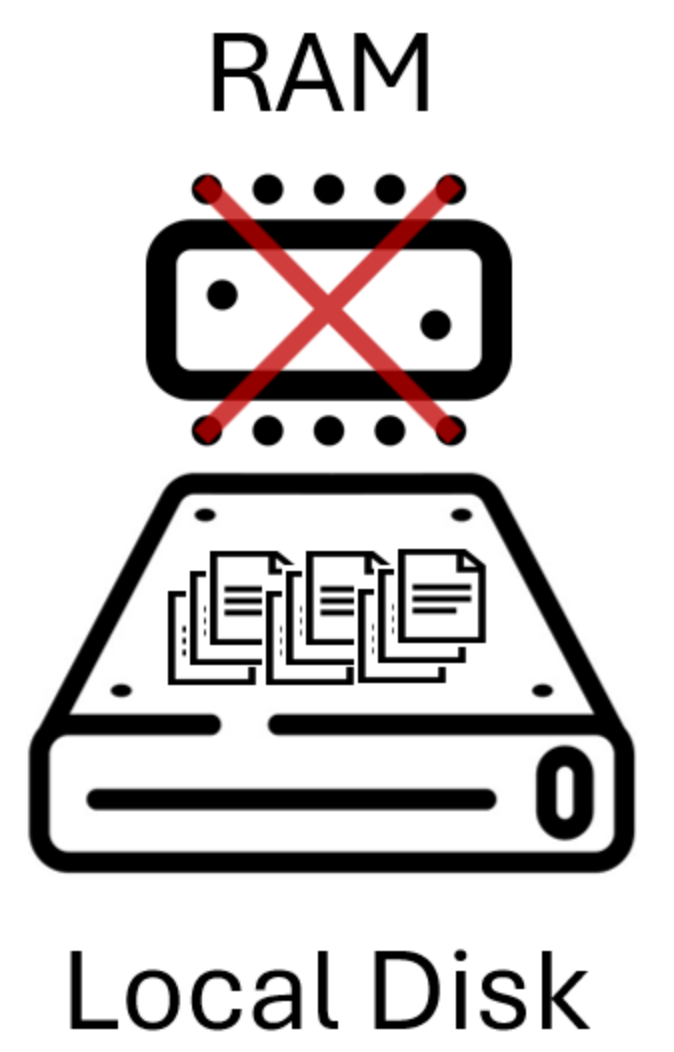
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Background

Vector Embeddings and Similarity Join

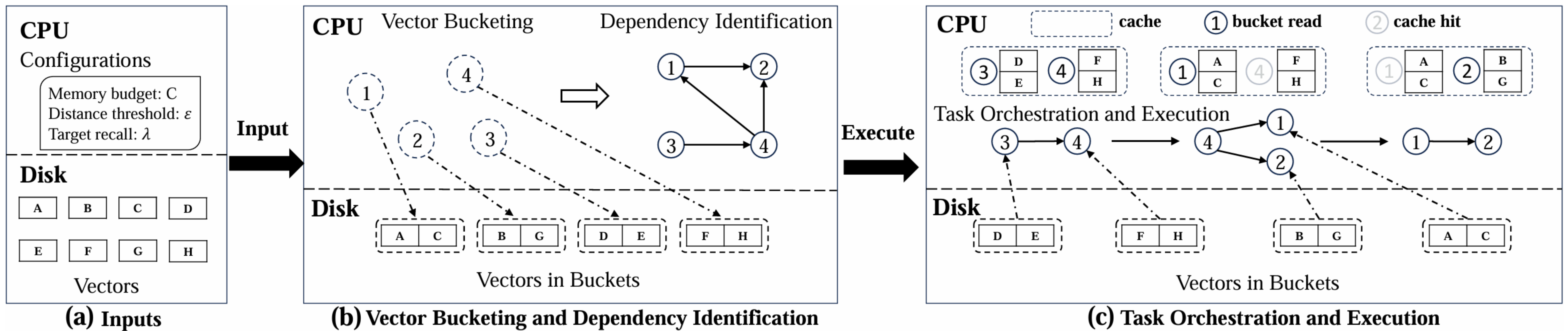


Demand for Scaling

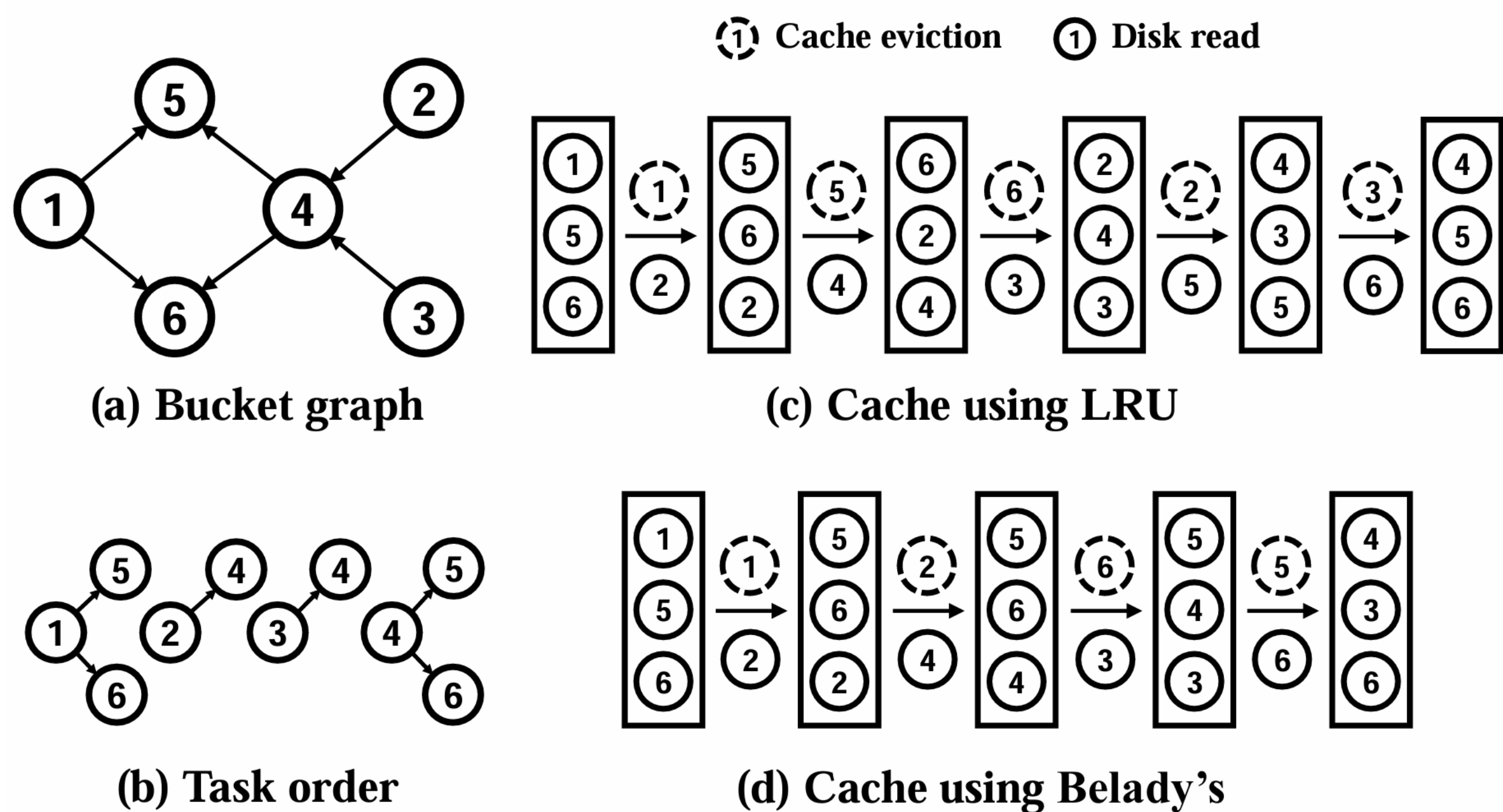


DiskJoin: A Disk-based Solution

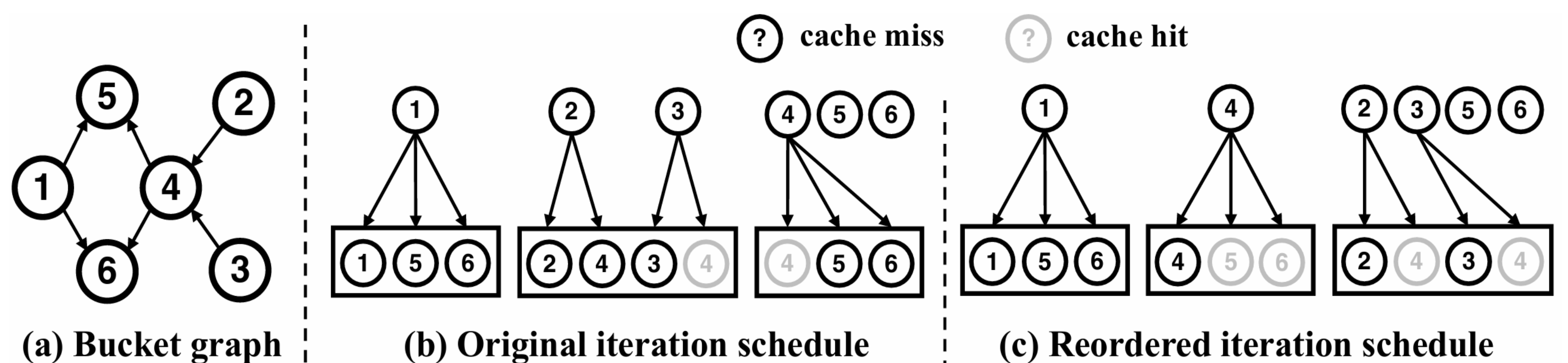
Workflow Overview



Optimal Cache Management



Task Ordering via Graph Reordering



Optimization Objective

$$F(P) = \sum_{0 < P(v) - P(u) \leq w} |\mathcal{N}(u) \cap \mathcal{N}(v)| = \sum_{i=1}^n \sum_{j=\max\{1, i-w\}}^{i-1} |\mathcal{N}(u) \cap \mathcal{N}(v)|$$

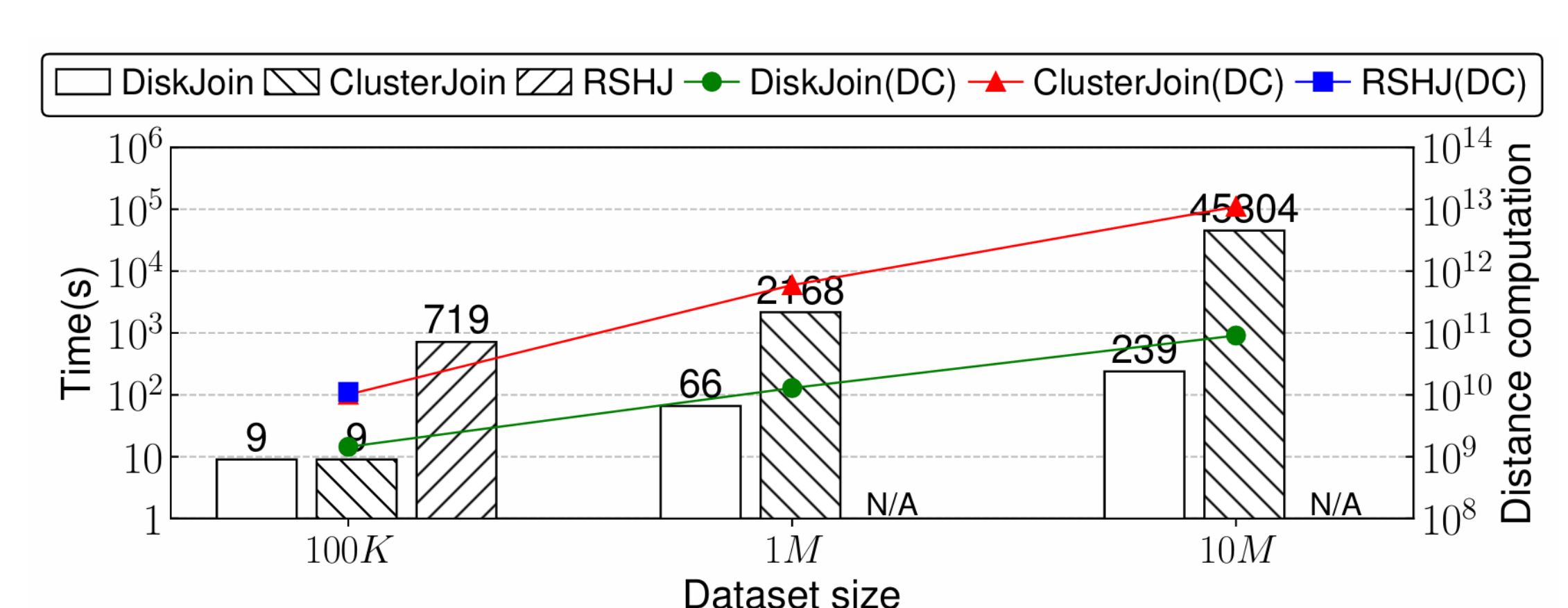
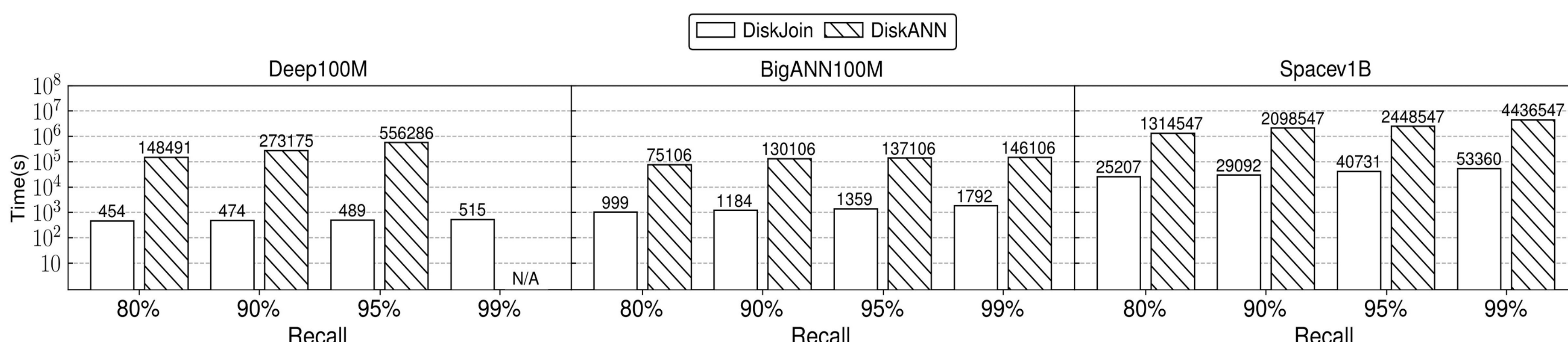
Greedy Node Selection

$$k_v = \sum_{j=\max\{1, i-w\}}^{i-1} |\mathcal{N}(P[j]) \cap \mathcal{N}(v)|$$

Other Optimizations

- Memory-efficient disk index construction
- HNSW index on bucket centers to accelerate candidate bucket identification

Evaluation



- DiskJoin outperforms distributed methods and DiskANN by orders of magnitude, on datasets of size up to 1.3 billion vectors
- DiskJoin successfully eliminates the disk access time bottleneck

